

Review retrospective:
*the quartic–sextic partition specialisation — two dead opens,
not one*

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Abstract

A soundness review of `Laplace/TwoD/QuarticSexticPartitionAsymptotic.lean`: the $(k_1, k_2) = (2, 3)$ specialisation of the 2D separable partition asymptotic, $Z_{2D}(t) = C'' \cdot t^{-1/4-1/6}$ for $L(x, y) = x^4/24 + y^6/720$. Result: **a clean fidelity pass with two dead-open removals** — both `open Real` and `open MeasureTheory`, the latter dead here (unlike its generic sibling) because the specialisation consumes the closed form directly and carries no integrability blocks.

1 Setting

This is the concrete $(2, 3)$ instance — the “first instance” per the seabed `CLAUDE.md` — of the partition leg whose generic- (k_1, k_2) form was reviewed at `KthKthPartitionAsymptotic`. It does not re-run the lift machinery: it takes the seabed closed form `partitionFunction_quarticSextic_eq` (from `TwoD/QuarticSextic.lean`) and repackages it as a `const·t(⋯)` identity, a rescaled `Tendsto`, and an `IsEquivalent`.

2 What was reviewed

Three public theorems plus the constant $C'' = \frac{1}{6} 24^{1/4} 720^{1/6} \Gamma(1/4) \Gamma(1/6)$. The constant is exactly the $(2, 3)$ instance of the generic `kthKthPartitionConst` ($4! = 24$, $6! = 720$, $\frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$), and the exponent $-1/4 - 1/6$ matches $-1/(2k_1) - 1/(2k_2)$.

3 Fidelity / soundness findings

Sound; no theorem-level mismatch. The constant and exponent are correct, the closed-form combination $(1/2)(1/3) = 1/6$ is right, and the exact theorem carries `ht : 0 < t` where needed. One report-only prose note: the module docstring states the closed-form identity without restating $t > 0$ — terse-but-true narrative (the t -power superscript carries the parameter, and the theorems require it), the same recurring elision seen across the `TwoD/` module docstrings; left as is, consistent with the `KthKthOddMoments` and `AddSeparable` reviews.

4 Simplifications applied

Two dead opens. `open Real` is dead (all `Real.*` qualified, bare t^{real} is `rpow`). And — the distinguishing feature of this file — `open MeasureTheory` is *also* dead: unlike the generic

sibling (and the numerator/moment legs), this specialisation has no `have hU/hV : Integrable ...` blocks, so there is no bare `Integrable` or `∫`; `partitionFunction/addSeparable` are used as project defs. With the open line reduced to `open Filter Asymptotics` (plus scoped `Topology`), the module is green at 2702 jobs, no warnings, signatures byte-identical. GPT affirmed (and had flagged the `MeasureTheory` removal as “if the build confirms” — it did).

5 Disagreements and open questions

None. The reviewer’s other suggestions (`show ... by ring → named-have`; dropping the terminal `exact` wrappers) were declined as proof-golf.

6 Lessons

- **A specialisation can be *more* trimmable than the generic it instantiates.** The generic legs keep `open MeasureTheory` load-bearing through their inline integrability proofs; this (2,3) file delegates the closed form to a seabed lemma and never touches `Integrable/∫`, so a second open dies. The dead-open verdict is per-file even within a copy-paste family — the build is the arbiter, and “if the build confirms” is the honest way to flag a second open.
- **The `t>0` module-prose elision is a family-wide pattern, not a per-file drift.** Several `TwoD/` module docstrings state the t -dependent identity without restating $t > 0$. Fixing them one at a time is noise; a single batched hygiene tide is the right granularity.

7 Follow-ups

- The remaining `QuarticSextic*` files (`Numerator/MomentAsymptotic`, `QuarticSextic`, `QuarticSexticMoments`, `QuarticSexticOddMoments`) — check each for the *second* dead open (`MeasureTheory`) as well as the first, since the asymptotic-packaging ones likely share this file’s no-integrability shape.
- Batched hygiene tide: add “for $t > 0$ ” to the `TwoD/` module docstrings that state a t -dependent closed form; plus the `show→have` proof-golf.